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Introduction

Cybersecurity is a critical concern for financial institutions such as banks, as they handle highly sensitive customer data, including account details and financial transactions. Due to the increasing number of cyber attacks targeting the banking sector, organizations must adopt a proactive approach to identifying risks and implementing effective security measures. This report analyses cybersecurity threats affecting a bank and proposes mitigation strategies while considering relevant legal and ethical requirements.

Threat Assessment and Vulnerabilities

Banks are primary targets for cybercriminals because of the financial value of the data they store. One of the most common threats is phishing, where attackers send fraudulent emails or messages to bank customers, attempting to steal login credentials. This can result in unauthorized access to bank accounts and financial loss.

In addition, banks are vulnerable to malware attacks, particularly ransomware, which can disrupt banking systems by encrypting critical data and demanding payment. Another major risk is unauthorized access caused by weak authentication systems.

These threats are often linked to specific vulnerabilities within the bank's systems. For example, the use of weak passwords without multi-factor authentication (MFA) makes customer accounts easier to compromise. Outdated banking software may also contain exploitable security flaws. Furthermore, insufficient network protection can expose financial data during online transactions. Human factors, such as lack of employee awareness, also increase the likelihood of successful attacks.

- **Mitigation Plan**

To mitigate these risks, banks should implement a layered security strategy. Firewalls should be deployed to monitor and control network traffic, preventing unauthorized access to internal banking systems. Additionally, multi-factor authentication (MFA) should be enforced for both customers and employees to strengthen account security.

Encryption plays a key role in protecting financial data. Secure protocols such as SSL/TLS should be used to protect online banking transactions, ensuring that sensitive information is not intercepted. Furthermore, all stored customer data should be encrypted to minimize the impact of potential data breaches.

Regular system updates and patch management are also essential to address known vulnerabilities in banking software. In addition, banks should provide cybersecurity training for employees to help them recognize phishing attempts and follow secure practices. This reduces the risk of human error, which is a common cause of security breaches in financial institutions.

Legal and Ethical Considerations

Banks must comply with data protection regulations such as GDPR, which require organizations to securely manage and protect customer data. Failure to comply can result in significant financial penalties and loss of customer trust. Additionally, banks should follow standards such as ISO/IEC 27001 to ensure a structured approach to information security management.

From an ethical perspective, banks have a responsibility to protect customer privacy and ensure confidentiality in all financial transactions. Transparency in how customer data is collected, stored, and used is essential to maintain trust and credibility.

Conclusion

In conclusion, banks face significant cybersecurity threats, including phishing, malware, and unauthorized access, which exploit both technical and human vulnerabilities. However, by implementing strong security measures such as firewalls, encryption, and multi-factor authentication, these risks can be effectively reduced. Compliance with legal frameworks and adherence to ethical standards are also essential in maintaining customer trust and ensuring long-term security.

References

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